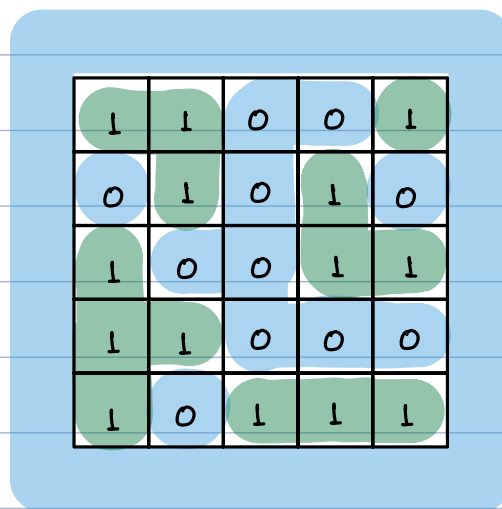


Number of Islands

You are given a 2D grid of '1's (land) and '0's (water). Your task is to determine the number of islands in the grid. An island is formed by connecting adjacent (horizontally or vertically) land cells. Diagonal connections are not considered.

Given here if the cell values has 1 then there is land and 0 if it is water, and you may assume all four edges of the grid are all surrounded by water.



No. of Islands = 5

Graph
Connected
Components
↓
DFS

```

int numIsland(grid [][]){
    m = grid.length;
    n = grid[0].length;

    c = 0;

    for (i = 0; i < m; i++) {
        for (j = 0; j < n; j++) {
            if (grid[i][j] == "1") {
                c++;
                dfs(grid, i, j);
            }
        }
    }

    return c;
}

```

```

void dfs(grid [][], int i, int j) {
    if (i < 0 || j < 0 || i >= grid.length || j >= grid[0].length ||
        grid[i][j] == '0') return;

    return;
}

```

```
grid[i][j] = '0';
```

```
dfs(grid, i+1, j);
```

```
dfs(grid, i-1, j);
```

```
dfs(grid, i, j+1);
```

```
dfs(grid, i, j-1);
```

```
}
```

TC: $O(N \times M)$

SC: $O(N \times M)$